

2(a)(i)	plant nurseries = (–) 69 (%) ; ;	2
2(a)(ii)	<i>descriptions of:</i> woodland and roadside, increase in number of cases, (over 2013–2015) or nurseries (and gardens) and farmland, decrease in number of cases (over 2013–2015) ; <i>explanations as ideas of:</i> 1 (more) difficult to manage, ‘wild’ environments / woodland / roadside, ora or 2 as woodland and roadside, cover a large area / far from centres of habitation, so, many trees to get infected / hard to get access for management difficult to access, ora or 3 disease spreads (more) easily, in, dense woodland, ora or 4 ref to burying diseased trees could provide a reservoir of infection or 5 <i>idea of</i> number of reported findings have increased as awareness of ash dieback has spread / AW ;	2

Question	Answer	Marks
2(b)(i)	complementary base pairing ;	1
2(b)(ii)	<i>idea of</i> intensity / brightness, of, fluorescence / fluorescent dye / colour or difference in intensity of, fluorescence / fluorescent dye / colour, (between healthy and diseased samples) ;	1
2(c)(i)	(these trees) are (likely to be) tolerant / resistant or (these trees could be) a source of, tolerance / resistance, genes / alleles / mRNA / DNA or <i>idea of</i> comparing non-diseased and diseased trees ;	1
2(c)(ii)	<i>any two of:</i> multiple / several / many / 4, locations ; varying ages ; from differently sized populations ; from, different countries / from across Europe ; varying degree of damage / healthy and non-healthy, sampled ; A description of a suitable, random / systematic, sampling technique for 1 mark <i>if no other mp awarded</i> .	2
2(c)(iii)	<i>any one of:</i> to allow (easier) comparison of data (from different sites) ; to allow (easier) analysis (of results) ; (easier) to plot, as graph / bar chart ; <i>idea of</i> (bioinformatic) analysis will be carried out by a computer ;	1
2(d)(i)	guanine / G ;	1
2(d)(ii)	<i>idea of</i> (mutation causes) a change in, protein / polypeptide / enzyme, which changes, function / tolerance / resistance or (mutation causes) a change in, protein / polypeptide / enzyme, that allows the plant to, respond to infection / repair damage or (mutation causes) a change in, protein / polypeptide / enzyme, that prevents the pathogen from, entering / spreading in, the ash plants ;	1